

```

Last login: Tue Jun  9 20:57:52 on ttys002
adityadave@Adityas-MacBook-Air-3 ~ % psql -d postgres
psql (18.4)
Type "help" for help.

postgres=# \c tpch
You are now connected to database "tpch" as user "adityadave".
tpch=# -- Step 1: Reset role and disable RLS
RESET ROLE;
ALTER TABLE orders DISABLE ROW LEVEL SECURITY;

-- Step 2: Drop existing view if any
DROP VIEW IF EXISTS orders_secure;

-- Step 3: Create view with same filter logic as RLS policy
CREATE VIEW orders_secure AS
SELECT o.*
FROM orders o
WHERE EXISTS (
    SELECT 1 FROM customer c
    WHERE c.c_custkey = o.o_custkey
    AND c.c_acctbal > 5000
);

-- Step 4: Grant permissions and set role
GRANT SELECT ON orders_secure TO alice;
GRANT SELECT ON customer TO alice;
GRANT SELECT ON lineitem TO alice;
SET ROLE alice;

-- Step 5: Run Q4 with timing
EXPLAIN (ANALYZE, VERBOSE, BUFFERS)
select
    o_orderpriority,
    count(*) as order_count
from
    orders_secure
where
    o_orderdate >= date '1994-01-01'
    and o_orderdate < date '1994-01-01' + interval '3' month
    and exists (
        select
            *
        from
            lineitem
        where
            l_orderkey = o_orderkey
            and l_commitdate < l_receiptdate
    )
group by
    o_orderpriority
order by
    o_orderpriority;
WARNING: ===== RAW PARSE TREE =====
DETAIL:  {RAWSTMT
:stmt
{VARIABLESETSTMT
:kind 4
:name role
:args <>
:jumble_args false
:is_local false
:location -1
}
:stmt_location 0
:stmt_len 10
}

WARNING: ===== ANALYZED QUERY =====
DETAIL:  {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{VARIABLESETSTMT
:kind 4
:name role
:args <>
:jumble_args false
:is_local false
:location -1
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:table <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 10
}

RESET
WARNING: ===== RAW PARSE TREE =====
DETAIL:  {RAWSTMT
:stmt
{ALTERTABLESTMT
:relation
{RANGEVAR
:catalogname <>
:schemaname <>
:relname orders
:inh true
:relpersistence p
:alias <>
:location 12
}
:cmds (
{ALTERTABLECMD
:subtype 55
:name <>
:num 0
:newowner <>
:def <>
:behavior 0

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        :missing_ok false
        :recurse false
    }
    :objtype 41
    :missing_ok false
}
:stmt_location 0
:stmt_len 45
}

WARNING: ===== ANALYZED QUERY =====
DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{ALTERTABLEMT
:relation
{RANGEVAR
:catalogname <>
:schemaname <>
:relname orders
:inh true
:relpersistence p
:alias <>
:location 12
}
:cmds {
{ALTERTABLECMD
:subtype 65
:name <>
:num 0
:newowner <>
:def <>
:behavior 0
:missing_ok false
:recurse false
}
}
:objtype 41
:missing_ok false
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable <>
:rtepermInfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 45
}

ALTER TABLE
WARNING: ===== RAW PARSE TREE =====
DETAIL: {RAWSTMT
:stmt
{DROPSMT
:objects {("orders_secure")}
:removeType 51
:behavior 0
:missing_ok true
:concurrent false
}
:stmt_location 0
:stmt_len 33
}

WARNING: ===== ANALYZED QUERY =====
DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{DROPSMT
:objects {("orders_secure")}
:removeType 51
:behavior 0
:missing_ok true
:concurrent false
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable <>
:rtepermInfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>

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:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 33
}

DROP VIEW
WARNING: ===== RAW PARSE TREE =====
DETAIL: {RAWSTMT
:stmt
{VIEWSTMT
:view
{RANGEVAR
:catalogname <>
:schemaname <>
:relname orders_secure
:inh true
:relpersistence p
:alias <>
:location 12
}
:aliases <>
:query
{SELECTSTMT
:distinctClause <>
:intoClause <>
:targetList (
{RETARGET
:name <>
:indirection <>
:val
{COLUMNREF
:fields ("o"
{A_STAR
}
)
:location 36
}
}
:location 36
)
}
:fromClause (
{RANGEVAR
:catalogname <>
:schemaname <>
:relname orders
:inh true
:relpersistence p
:alias
{ALIAS
:aliasname o
:colnames <>
}
:location 45
}
)
:whereClause
{SUBLINK
:sublinkType 0
:sublinkId 0
:testexpr <>
:operName <>
:subselect
{SELECTSTMT
:distinctClause <>
:intoClause <>
:targetList (
{RETARGET
:name <>
:indirection <>
:val
{A_CONST
:val 1
:location 80
}
:location 80
}
)
:fromClause (
{RANGEVAR
:catalogname <>
:schemaname <>
:relname customer
:inh true
:relpersistence p
:alias
{ALIAS
:aliasname c
:colnames <>
}
:location 87
}
)
:whereClause
{BOOLEXP
:boolop and
:args (
{A_EXPR
:name ("=")
:lexpr
{COLUMNREF
:fields ("c" "c_custkey")
:location 100
}
:rexpr
{COLUMNREF
:fields ("o" "o_custkey")
:location 122
}
}
:rexpr_list_start 0
:rexpr_list_end 0
:location 120
}
{A_EXPR
:name (">")
:lexpr
{COLUMNREF
:fields ("c" "c_acctbal")
:location 142
}
:rexpr
{A_CONST
:val 5000
:location 156
}
:rexpr_list_start 0
:rexpr_list_end 0
:location 154
}
)
:location 138
}
:groupClause <>
:groupDistinct false
:havingClause <>
>windowClause <>
:valuesLists <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:lockingClause <>
:withClause <>
:op 0
:all false

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        :larg <>
        :rarg <>
    }
    :location 60
}
:groupClause <>
:groupDistinct false
:havingClause <>
>windowClause <>
:valuesLists <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:lockingClause <>
:withClause <>
:op 0
:all false
:larg <>
:rarg <>
}
:replace false
:options <>
:withCheckOption 0
}
:stmt_location 0
:stmt_len 162
}

WARNING: ===== ANALYZED QUERY =====
DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{VIEWSTMT
:view
{RANGEVAR
:catalogname <>
:schemaname <>
:relname orders_secure
:inh true
:relpersistence p
:alias <>
:location 12
}
}
:aliases <>
:query
{SELECTSTMT
:distinctClause <>
:intoClause <>
:targetList (
{REXTARGET
:name <>
:indirection <>
:val
{COLUMNREF
:fields ("o"
{A_STAR
}
}
:location 36
}
:location 36
}
)
:fromClause (
{RANGEVAR
:catalogname <>
:schemaname <>
:relname orders
:inh true
:relpersistence p
:alias
{ALIAS
:aliasname o
:colnames <>
}
:location 45
}
)
:whereClause
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{SELECTSTMT
:distinctClause <>
:intoClause <>
:targetList (
{REXTARGET
:name <>
:indirection <>
:val
{A_CONST
:val 1
:location 80
}
:location 80
}
)
:fromClause (
{RANGEVAR
:catalogname <>
:schemaname <>
:relname customer
:inh true
:relpersistence p
:alias
{ALIAS
:aliasname c
:colnames <>
}
:location 87
}
)
:whereClause
{BOOLEXP
:subloop and
:args (
{A_EXPR
:name ("=")
:lexpr
{COLUMNREF
:fields ("c" "c_custkey")
:location 108
}
:rexpr
{COLUMNREF
:fields ("o" "o_custkey")
:location 122
}
}
:rexpr_list_start 0
:rexpr_list_end 0
:location 120
}
{A_EXPR
:name (">")
:lexpr
{COLUMNREF
:fields ("c" "c_acctbal")
:location 142
}
:rexpr
{A_CONST
:val 5000

```

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        :location 156
      }
      :rexpr_list_start 0
      :rexpr_list_end 0
      :location 154
    }
  )
  :location 138
}
:groupClause <>
:groupDistinct false
:havingClause <>
>windowClause <>
:valuesLists <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:lockingClause <>
:withClause <>
:op 0
:all false
:larg <>
:rarg <>
}
:location 60
}
:groupClause <>
:groupDistinct false
:havingClause <>
>windowClause <>
:valuesLists <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:lockingClause <>
:withClause <>
:op 0
:all false
:larg <>
:rarg <>
}
:replace false
:options <>
:withCheckOption 0
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 162
}

```

WARNING: ===== ANALYZED QUERY =====

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DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTY
:alias
  {ALIAS
:aliasname o
:colnames <>
}
:eref
  {ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 16561
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
  {RTEPERMISSIONINFO
:relid 16561
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
  {FROMEXPR
:fromlist (

```

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{RANGETBLREF
:rtindex 1
}
)
:quals
{SUBLINK
:sublinkType 0
:sublinkId 0
:testexpr <>
:opetName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rttable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname c
:colnames <>
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c
_phone" "c_acctbal" "c_mktsegment" "c_comment")
}
:rtkind 0
:relid 16546
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesampl <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rtperminfos (
{RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromList (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:varitypm -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 108
}
)
{VAR
:varno 1
:varattno 2
:vartype 23
:varitypm -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 122
}
)
:location 120
}
)
{OPEXP
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 701
:varitypm -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 142
}
)
{FUNCEXP
:funcid 316
:funcresulttype 701
:funcset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:constitypm -1
:constcollid 0
:constlen 4
}
)
}
}
}

```

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        :constbyval true
        :constisnull false
        :location 156
        :constvalue 4 [ -128 19 0 0 0 0 0 0 ]
    }
}
:location -1
}
}
:location 154
}
}
:location 138
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 80
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 60
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 36
}
:resno 1
:resname o_orderkey
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 36
}
:resno 2
:resname o_custkey
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 36
}
:resno 3
:resname o_orderstatus
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 36
}

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    }
:resno 4
:resname o_totalprice
:ressortgrouppref 0
:resorigtbl 16561
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 36
}
:resno 5
:resname o_orderdate
:ressortgrouppref 0
:resorigtbl 16561
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 36
}
:resno 6
:resname o_orderpriority
:ressortgrouppref 0
:resorigtbl 16561
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location 36
}
:resno 7
:resname o_clerk
:ressortgrouppref 0
:resorigtbl 16561
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location 36
}
:resno 8
:resname o_shippriority
:ressortgrouppref 0
:resorigtbl 16561
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location 36
}
:resno 9
:resname o_comment
:ressortgrouppref 0
:resorigtbl 16561
:resorigcol 9
:resjunk false
}
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 162
}
}

CREATE VIEW
WARNING: ===== RAW PARSE TREE =====
DETAIL: {RAWSTMT
:stmt
{GRANTSTMT
:is_grant true

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```

:targettype 0
:objtype 41
:objects (
  {RANGEVAR
    :catalogname <>
    :schemaname <>
    :relname orders_secure
    :inh true
    :relpersistence p
    :alias <>
    :location 16
  }
)
:privileges (
  {ACCESSPRIV
    :priv_name select
    :cols <>
  }
)
:grantees (
  {ROLESPEC
    :roletype 0
    :rolename alice
    :location 33
  }
)
:grant_option false
:grantor <>
:behavior 0
}
:stmt_location 0
:stmt_len 38
}

WARNING: ===== ANALYZED QUERY =====
DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{GRANTSTMT
:is_grant true
:targettype 0
:objtype 41
:objects (
  {RANGEVAR
    :catalogname <>
    :schemaname <>
    :relname orders_secure
    :inh true
    :relpersistence p
    :alias <>
    :location 16
  }
)
:privileges (
  {ACCESSPRIV
    :priv_name select
    :cols <>
  }
)
:grantees (
  {ROLESPEC
    :roletype 0
    :rolename alice
    :location 33
  }
)
:grant_option false
:grantor <>
:behavior 0
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 38
}

GRANT
WARNING: ===== RAW PARSE TREE =====
DETAIL: {RAWSTMT
:stmt
{GRANTSTMT
:is_grant true
:targettype 0
:objtype 41
:objects (
  {RANGEVAR
    :catalogname <>
    :schemaname <>
    :relname customer
    :inh true
    :relpersistence p
    :alias <>
    :location 16
  }
)
:privileges (
  {ACCESSPRIV
    :priv_name select
    :cols <>
  }
)
:grantees (
  {ROLESPEC
    :roletype 0
    :rolename alice
    :location 28
  }
)
:grant_option false

```

```
      :grantor <>
      :behavior 0
    }
  :stmt_location 0
  :stmt_len 33
}
```

WARNING: ===== ANALYZED QUERY =====

```
DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{GRANTSTMT
:is_grant true
:targetType 0
:objtype 41
:objects (
{RANGEVAR
:catalogname <>
:schemaname <>
:relname customer
:inh true
:relpersistence p
:alias <>
:location 16
}
)
:privileges (
{ACCESSPRIV
:priv_name select
:cols <>
}
)
:grantees (
{ROLESPEC
:roletype 0
:rolename alice
:location 28
}
)
:grant_option false
:grantor <>
:behavior 0
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable <>
:rteperinfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintOps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 33
}
```

GRANT
WARNING: ===== RAW PARSE TREE =====

```
DETAIL: {RAWSTMT
:stmt
{GRANTSTMT
:is_grant true
:targetType 0
:objtype 41
:objects (
{RANGEVAR
:catalogname <>
:schemaname <>
:relname lineitem
:inh true
:relpersistence p
:alias <>
:location 16
}
)
:privileges (
{ACCESSPRIV
:priv_name select
:cols <>
}
)
:grantees (
{ROLESPEC
:roletype 0
:rolename alice
:location 28
}
)
:grant_option false
:grantor <>
:behavior 0
}
:stmt_location 0
:stmt_len 33
}
```

WARNING: ===== ANALYZED QUERY =====

```
DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{GRANTSTMT
:is_grant true
:targetType 0
:objtype 41
:objects (
{RANGEVAR
:catalogname <>
:schemaname <>
:relname lineitem
:inh true
:relpersistence p
:alias <>
:location 16
}
)
```

```

    )
    :privileges (
    {ACCESSPRIV
    :priv_name select
    :cols <>
    }
    )
    :grantees (
    {ROLESPEC
    :roletype 0
    :rolename alice
    :location 28
    }
    )
    :grant_option false
    :grantor <>
    :behavior 0
    }
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 33
}

GRANT
WARNING: ===== RAW PARSE TREE =====
DETAIL: {RAWSTMT
:stmt
{VARIABLESETSTMT
:kind 0
:name role
:args (
{A_CONST
:val "alice"
:location 9
}
)
:jumble_args false
:is_local false
:location 9
}
:stmt_location 0
:stmt_len 14
}

WARNING: ===== ANALYZED QUERY =====
DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{VARIABLESETSTMT
:kind 0
:name role
:args (
{A_CONST
:val "alice"
:location 9
}
)
:jumble_args false
:is_local false
:location 9
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 14
}

SET
WARNING: ===== RAW PARSE TREE =====
DETAIL: {RAWSTMT
:stmt

```

```

{EXPLAINSTMT
:query
{SELECTSTMT
:distinctClause <>
:intoClause <>
:targetList (
{RESTARTGET
:name <>
:indirection <>
:val
{COLUMNREF
:fields ("o_orderpriority")
:location 47
}
:location 47
}
{RESTARTGET
:name order_count
:indirection <>
:val
{FUNCCALL
:funcname ("count")
:args <>
:agg_order <>
:agg_filter <>
:over <>
:agg_within_group false
:agg_star true
:agg_distinct false
:func_variadic false
:funcformat 0
:location 68
}
:location 68
}
)
:fromClause (
{RANGEVAR
:catalogname <>
:schemaname <>
:relname orders_secure
:inh true
:relpersistency p
:alias <>
:location 101
}
)
:whereClause
{BOOLEXPRESS
:boolop and
:args (
{A_EXPR
:name (">=")
:lexpr
{COLUMNREF
:fields ("o_orderdate")
:location 125
}
}
:rexpr
{TYPECAST
:arg
{A_CONST
:val "\1994-01-01"
:location 145
}
}
:typeName
{TYPPNAME
:names ("date")
:typeid 0
:setof false
:pct_type false
:typmods <>
:typemod -1
:arrayBounds <>
:location 140
}
:location -1
}
:rexpr_list_start 0
:rexpr_list_end 0
:location 137
}
{A_EXPR
:name ("<")
:lexpr
{COLUMNREF
:fields ("o_orderdate")
:location 166
}
}
:rexpr
{A_EXPR
:name ("+")
:lexpr
{TYPECAST
:arg
{A_CONST
:val "\1994-01-01"
:location 185
}
}
:typeName
{TYPPNAME
:names ("date")
:typeid 0
:setof false
:pct_type false
:typmods <>
:typemod -1
:arrayBounds <>
:location 180
}
:location -1
}
:rexpr
{TYPECAST
:arg
{A_CONST
:val "\3"
:location 209
}
}
:typeName
{TYPPNAME
:names ("pg_catalog" "interval")
:typeid 0
:setof false
:pct_type false
:typmods (
{A_CONST
:val 2
:location 213
}
)
:typemod -1
:arrayBounds <>
:location 200
}
:location -1
}
:rexpr_list_start 0
:rexpr_list_end 0
:location 198
}
:rexpr_list_start 0
:rexpr_list_end 0
:location 178
}
{SUBLINK
:sublinkType 0
:subLinkId 0

```

```

:texexpr <>
:operName <>
:subselect
{SELECTSTMT
:distinctClause <>
:intoClause <>
:targetList (
{RESTARTGET
:name <>
:indirection <>
:val
{COLUMNREF
:fields (
{A_STAR
}
}
:location 263
}
:location 263
}
)
:fromClause (
{RANGEVAR
:catalogname <>
:schemaname <>
:relname lineitem
:inh true
:relpersistence p
:alias <>
:location 290
}
)
:whereClause
{BOOLEXP
:boolop and
:args (
{A_EXPR
:name ("=")
:lexpr
{COLUMNREF
:fields ("l_orderkey")
:location 325
}
}
:rexpr
{COLUMNREF
:fields ("o_orderkey")
:location 338
}
}
:rexpr_list_start 0
:rexpr_list_end 0
:location 336
}
{A_EXPR
:name ("<")
:lexpr
{COLUMNREF
:fields ("l_commitdate")
:location 365
}
}
:rexpr
{COLUMNREF
:fields ("l_receiptdate")
:location 380
}
}
:rexpr_list_start 0
:rexpr_list_end 0
:location 378
}
}
:location 361
}
:groupClause <>
:groupDistinct false
:havingClause <>
>windowClause <>
:valuesLists <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:lockingClause <>
:withClause <>
:op 0
:all false
:larg <>
:rarg <>
}
:location 227
}
)
:location 162
}
:groupClause (
{COLUMNREF
:fields ("o_orderpriority")
:location 413
}
)
:groupDistinct false
:havingClause <>
>windowClause <>
:valuesLists <>
:sortClause (
{SORTBY
:node
{COLUMNREF
:fields ("o_orderpriority")
:location 442
}
:sortby_dir 0
:sortby_nulls 0
:useOp <>
:location -1
}
)
:limitOffset <>
:limitCount <>
:limitOption 0
:lockingClause <>
:withClause <>
:op 0
:all false
:larg <>
:rarg <>
}
:options (
{DEFELEM
:defnamespace <>
:defname analyze
:arg <>
:defaction 0
:location 9
}
{DEFELEM
:defnamespace <>
:defname verbose
:arg <>
:defaction 0
:location 18
}
{DEFELEM
:defnamespace <>
:defname buffers
:arg <>
:defaction 0
:location 27
}
)
)

```

```

    }
    :stmt_location 0
    :stmt_len 457
    }
}

WARNING: ===== ANALYZED QUERY =====
DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
:utilityStmt
{EXPLAINSTMT
:query
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:ctelist <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname orders_secure
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalpri
ce" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority"
"o_comment")
}
:rtekind 0
:relid 16839
:inh true
:relkind v
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCI true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
:rtekind 9
:groupexprs (
{VAR
:varno 1
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 47
}
)
:lateral false
:inFromCI false
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16839
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 125
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 145
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
)
:location 137
}
{OPEXP
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:vartype 1082

```

```

:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 166
}
{OPEXPR
:opno 1076
:opfuncid 2071
:opresulttype 1114
:opretset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 185
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
{CONST
:consttype 1186
:consttypmod 196607
:constcollid 0
:constlen 16
:constbyval false
:constisnull false
:location 209
:constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]
}
)
:location 198
}
)
:location 178
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBLENTTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_
linenumber" "l_quantity" "l_extendedprice" "l_disco
unt" "l_tax" "l_returnflag" "l_linestatus" "l_shipd
ate" "l_commitdate" "l_receiptdate" "l_shipinstruct
" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20
21 22 23)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXPR
:boolop and
:args (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 325
}
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 338
}
}
}
}

```

```

    )
    :location 336
  }
  {OEXPR
  :opno 1095
  :opfuncid 1087
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
  {VAR
  :varno 1
  :varattno 12
  :vartype 1082
  :vartypmod -1
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 1
  :varattnosyn 12
  :location 365
  }
  {VAR
  :varno 1
  :varattno 13
  :vartype 1082
  :vartypmod -1
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 1
  :varattnosyn 13
  :location 380
  }
  )
  :location 378
  }
  )
  :location 361
  }
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 263
}
}
:resno 1
:resname l_orderkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 263
}
}
:resno 2
:resname l_partkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 263
}
}
:resno 3
:resname l_suppkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 263
}
}
:resno 4
:resname l_linenummer
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1

```

```

:varattnosyn 5
:location 263
}
}
:resno 5
:resname l_quantity
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 263
}
}
:resno 6
:resname l_extendedprice
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location 263
}
}
:resno 7
:resname l_discount
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location 263
}
}
:resno 8
:resname l_tax
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location 263
}
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location 263
}
}
:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 263
}
}
:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082

```

```

        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 12
        :location 263
    }
    :resno 12
    :resname l_commitdate
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 12
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 13
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 13
    :location 263
    }
    :resno 13
    :resname l_receiptdate
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 13
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 14
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 14
    :location 263
    }
    :resno 14
    :resname l_shipinstruct
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 14
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 15
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 15
    :location 263
    }
    :resno 15
    :resname l_shipmode
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 15
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 16
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 16
    :location 263
    }
    :resno 16
    :resname l_comment
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 16
    :resjunk false
    }
    }
    :location 227
    }
    }
    :location 162
    }
    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList (
    {TARGETENTRY
    :expr
    {VAR
    :varno 2
    :varattno 1
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0

```

```

:varnosyn 2
:varattnosyn 1
:location -1
}
:resno 1
:resname o_orderpriority
:resortgroupref 1
:resorigtbl 16839
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2883
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 0
:aggno -1
:aggtransno -1
:location 68
}
:resno 2
:resname order_count
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
{SORTGROUPCLAUSE
:t1SortGroupRef 1
:eqop 98
:sortop 664
:reverse_sort false
:nulls_first false
:hashable true
}
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
{SORTGROUPCLAUSE
:t1SortGroupRef 1
:eqop 98
:sortop 664
:reverse_sort false
:nulls_first false
:hashable true
}
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:options (
{DEFELEM
:defnamespace <>
:define analyze
:arg <>
:deffaction 0
:location 9
}
{DEFELEM
:defnamespace <>
:define verbose
:arg <>
:deffaction 0
:location 18
}
{DEFELEM
:defnamespace <>
:define buffers
:arg <>
:deffaction 0
:location 27
}
}
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 457

```

```

}

WARNING: ===== REWRITE INPUT =====
DETAIL:  QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilitySmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname orders_secure
        :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
      }
    :rtekind 0
    :relid 16839
    :inh true
    :relkind v
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname *GROUP*
        :colnames ("o_orderpriority")
      }
    :rtekind 9
    :groupexprs (
      {VAR
        :varno 1
        :varattno 6
        :vartype 25
        :vartypmod -1
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 6
        :location 47
      }
    )
    :lateral false
    :inFromCl false
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 16839
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 12 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
  )
}
:quals
{BOOLEXP
  :boolop and
  :args (
    {OPEXP
      :opno 1098
      :opfuncid 1090
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
          :varno 1
          :varattno 5
          :vartype 1082
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 5
          :location 125
        }
      )
      {CONST
        :consttype 1082
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location 145
        :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
      }
    }
  )
  :location 137
}
{OPEXP
  :opno 2345
  :opfuncid 2338
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
      :varno 1
      :varattno 5
      :vartype 1082
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 5
      :location 166
    }
  )
}
{OPEXP
  :opno 1076

```

```

:opfuncid 2071
:opresulttype 1114
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {CONST
    :consttype 1082
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location 185
    :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
  }
  {CONST
    :consttype 1186
    :consttypmod 196607
    :constcollid 0
    :constlen 16
    :constbyval false
    :constisnull false
    :location 209
    :constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]
  }
)
:location 198
}
}
:location 178
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
  {RANGETBLENTRY
    :alias <>
    :eref
    {ALIAS
      :aliasname lineitem
      :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenu
        mber" "l_quantity" "l_extendedprice" "l_discount" "l_tax"
        "l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate
        " "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_commen
        t")
    }
    :rtekind 0
    :relid 16566
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 16566
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
      23)
    :insertedCols (b)
    :updatedCols (b)
  }
)
}
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
    :rtindex 1
  }
)
}
:quals
{BOOLEXP
:boolop and
:args (
  {OPEXP
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location 325
      }
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 1
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location 338
      }
    )
  }
:location 336
}
{OPEXP
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (

```

```

        {VAR
        :varno 1
        :varattno 12
        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 12
        :location 365
        }
        {VAR
        :varno 1
        :varattno 13
        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 13
        :location 380
        }
    )
    :location 378
}
}
:location 361
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 263
}
:resno 1
:resname l_orderkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 263
}
:resno 2
:resname l_partkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 263
}
:resno 3
:resname l_suppkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 263
}
:resno 4
:resname l_linenum
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 263
}
:resno 5
:resname l_quantity
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 5
:resjunk false
}
{TARGETENTRY

```

```

:expr
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 263
}
:resno 6
:resname l_extendedprice
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location 263
}
:resno 7
:resname l_discount
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location 263
}
:resno 8
:resname l_tax
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location 263
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location 263
}
:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 263
}
:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 263
}
:resno 12
:resname l_commitdate

```

```

:ressortgroupref 0
:resorigtbl 16566
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 263
}
:resno 13
:resname 1_receiptdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location 263
}
:resno 14
:resname 1_shipinstruct
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location 263
}
:resno 15
:resname 1_shipmode
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 15
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 16
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 16
:location 263
}
:resno 16
:resname 1_comment
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 16
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 227
}
}
:location 162
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
:resno 1
:resname o_orderpriority
:ressortgroupref 1
:resorigtbl 16839
:resorigcol 6
:resjunk false
}

```

```

{TARGETENTRY
:expr
  {AGGREF
  :aggnoid 2803
  :aggtype 20
  :aggcollid 0
  :inputcollid 0
  :aggtranstype 0
  :aggargtypes <>
  :aggdirectargs <>
  :args <>
  :aggorder <>
  :aggdistinct <>
  :aggfilter <>
  :aggstar true
  :aggvariadic false
  :aggkind n
  :aggpresorted false
  :agglevelsup 0
  :aggsplit 0
  :aggnr -1
  :aggtransno -1
  :location 68
  }
:resno 2
:resname order_count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
  {SORTGROUPCLAUSE
  :tleSortGroupRef 1
  :eqop 98
  :sortop 664
  :reverse_sort false
  :nulls_first false
  :hashable true
  }
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
  {SORTGROUPCLAUSE
  :tleSortGroupRef 1
  :eqop 98
  :sortop 664
  :reverse_sort false
  :nulls_first false
  :hashable true
  }
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

WARNING: ===== REWRITE OUTPUT =====

```

DETAIL: (
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:cteList <>
:rtable (
  {RANGEABLEENTRY
  :alias <>
  :eref
  {ALIAS
  :aliasname orders_secure
  :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
  }
}
:rtkind 1
:subquery
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGEABLEENTRY
  :alias
  {ALIAS
  :aliasname o
  :colnames <>
  }
}
:eref
{ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
}
:rtkind 0
:relid 16561
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
)

```

```

:rteperminfos (
  {RTEPERMISSIONINFO
  :relid 16561
  :inh true
  :requiredPerms 2
  :checkAsUser 10
  :selectedCols (b 8 9 10 11 12 13 14 15 16)
  :insertedCols (b)
  :updatedCols (b)
  }
)
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
    :rtindex 1
    }
  )
  :quals
  {SUBLINK
  :subLinkType 0
  :subLinkId 0
  :testexpr <>
  :operName <>
  :subselect
  {QUERY
  :commandType 1
  :querySource 0
  :canSetTag true
  :utilityStmt <>
  :resultRelation 0
  :hasAggs false
  :hasWindowFuncs false
  :hasTargetSRFs false
  :hasSublinks false
  :hasDistinctOn false
  :hasRecursive false
  :hasModifyingCTE false
  :hasForUpdate false
  :hasRowSecurity false
  :hasGroupRTE false
  :isReturn false
  :cteList <>
  :rtable (
    {RANGETBLENTRY
    :alias
    {ALIAS
    :aliasname c
    :colnames <>
    }
    :eref
    {ALIAS
    :aliasname c
    :colnames ("c_custkey" "c_name" "c_address" "c_nationk
ey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")
    }
    :rtekind 0
    :relid 16546
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
  )
  :rteperminfos (
    {RTEPERMISSIONINFO
    :relid 16546
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 13)
    :insertedCols (b)
    :updatedCols (b)
    }
  )
}
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
    :rtindex 1
    }
  )
  :quals
  {BOOLEXP
  :boolop and
  :args (
    {OPEXP
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location -1
      }
      {VAR
      :varno 1
      :varattno 2
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 1
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 2
      :location -1
      }
    )
    :location -1
  }
  {OPEXP
  :opno 674
  :opfuncid 297
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
    :varno 1
    :varattno 6
    :vartype 701
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 6
    :location -1
    }
  )
}

```

```

{FUNCEXP
:funcid 316
:funcresulttype 701
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
  {CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ -120 19 0 0 0 0 0 0 ]
}
)
:location -1
}
)
:location -1
}
)
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
:expr
  {CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
:expr
  {VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname o_orderkey
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 1
:resjunk false
}
  {TARGETENTRY
:expr
  {VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname o_custkey
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 2
:resjunk false
}
  {TARGETENTRY
:expr
  {VAR
:varno 1
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname o_orderstatus
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 3
:resjunk false
}
)

```

```

{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:varatype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname o_totalprice
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:varatype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname o_orderdate
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:varatype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname o_orderpriority
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:varatype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname o_clerk
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
:resno 8
:resname o_shippriority
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:varatype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
:resno 9
:resname o_comment
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 9
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
:windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0

```

```

:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier false
:relid 16839
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
:rtekind 9
:groupexprs (
{VAR
:varno 1
:varattno 6
:varatype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 47
}
)
:lateral false
:inFromCl false
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16839
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varatype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 125
}
)
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 145
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
)
:location 137
}
{OPEXP
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varatype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 166
}
)
{OPEXP
:opno 1076
:opfuncid 2071
:opresulttype 1114
:opretset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 185
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
)
{CONST
:consttype 1186
:consttypmod 196007
:constcollid 0
:constlen 16
:constbyval false
:constisnull false
:location 209
}
}
}

```

```

        :constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]
    }
}
:location 198
}
)
:location 178
}
{SUBLINK
:sublinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenu
mber" "l_quantity" "l_extendedprice" "l_discount" "l_tax"
"l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate
" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_commen
t")
}
:rtekind 0
:relid 16566
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromC1 true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
23)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 325
}
)
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 338
}
)
}
:location 336
}
)
{OPEXP
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 365
}
)
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13

```

```

        :location 380
    }
}
:location 378
}
)
:location 361
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 263
}
:resno 1
:resname l_orderkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 263
}
:resno 2
:resname l_partkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 263
}
:resno 3
:resname l_suppkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 263
}
:resno 4
:resname l_linenum
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 263
}
:resno 5
:resname l_quantity
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 263
}
:resno 6
:resname l_extendedprice
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 6
:resjunk false
}
}
{TARGETENTRY
:expr
{VAR

```

```

:varno 1
:varattno 7
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location 263
}
:resno 7
:resname l_discount
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location 263
}
:resno 8
:resname l_tax
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location 263
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location 263
}
:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 263
}
:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 263
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 263
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 16566

```

```

:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location 263
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location 263
}
:resno 15
:resname l_shipmode
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 15
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 16
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 16
:location 263
}
:resno 16
:resname l_comment
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 16
:resjunk false
}
}
:location 227
}
}
:location 162
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
:resno 1
:resname o_orderpriority
:ressortgroupref 1
:resorigtbl 16839
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 0
:aggno -1
:aggtransno -1
:location 68
}
}

```

```

:resno 2
:resname order_count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
)

```

WARNING: ===== PLANNER INPUT =====

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTY
    :alias <>
    :eref
      {ALIAS
        :aliasname orders_secure
        :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
      }
    :rtekind 1
    :subquery
      {QUERY
        :commandType 1
        :querySource 0
        :canSetTag true
        :utilityStmt <>
        :resultRelation 0
        :hasAggs false
        :hasWindowFuncs false
        :hasTargetSRFs false
        :hasSubLinks true
        :hasDistinctOn false
        :hasRecursive false
        :hasModifyingCTE false
        :hasForUpdate false
        :hasRowSecurity false
        :hasGroupRTE false
        :isReturn false
        :cteList <>
        :rtable (
          {RANGETBLENTY
            :alias
              {ALIAS
                :aliasname o
                :colnames <>
              }
            :eref
              {ALIAS
                :aliasname o
                :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
              }
            :rtekind 0
            :relid 16561
            :inh true
            :relkind F
            :rellockmode 1
            :perminfoindex 1
            :tablesample <>
            :lateral false
            :inFromCl true
            :securityQuals <>
          }
        )
        :rteperminfos (
          {RTEPERMISSIONINFO
            :relid 16561
            :inh true
            :requiredPerms 2
            :checkasUser 10
            :selectedCols (b 8 9 10 11 12 13 14 15 16)
            :insertedCols (b)
            :updatedCols (b)
          }
        )
      }
    :jointree
      {FROMEXPR
        :fromlist (
          {RANGETBLREF
            :rtindex 1
          }
        )
      }
    :quals
      {SUBLINK
        :sublinkType 0
        :subLinkId 0
        :testexpr <>
        :opername <>
      }
    }
}

```

```

:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname c
:colnames <>
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationk
ey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")
}
:rtekind 0
:relid 16546
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:joinTree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
}
:location -1
}
{OPEXP
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{FUNCEXP
:funcid 316
:funcresulttype 701
:funcset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ -120 19 0 0 0 0 0 0 ]
}
}
:location -1
}
)
:location -1
}

```

```

    }
    }
    :location -1
  }
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location -1
      :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
  :override 0
  :onConflict <>
  :returningOldAlias <>
  :returningNewAlias <>
  :returningList <>
  :groupClause <>
  :groupDistinct false
  :groupingSets <>
  :havingQual <>
  :windowClause <>
  :distinctClause <>
  :sortClause <>
  :limitOffset <>
  :limitCount <>
  :limitOption 0
  :rowMarks <>
  :setOperations <>
  :constraintDeps <>
  :withCheckOptions <>
  :stmt_location -1
  :stmt_len -1
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname o_orderkey
    :ressortgroupref 0
    :resorigtbl 16561
    :resorigcol 1
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 2
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 2
      :location -1
    }
    :resno 2
    :resname o_custkey
    :ressortgroupref 0
    :resorigtbl 16561
    :resorigcol 2
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 3
      :vartype 25
      :vartypmod -1
      :varcollid 100
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 3
      :location -1
    }
    :resno 3
    :resname o_orderstatus
    :ressortgroupref 0
    :resorigtbl 16561
    :resorigcol 3
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 4
      :vartype 701
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 4
      :location -1
    }
    :resno 4
    :resname o_totalprice
    :ressortgroupref 0
    :resorigtbl 16561
    :resorigcol 4
    :resjunk false
  }
  {TARGETENTRY
    :expr

```

```

{VAR
:varno 1
:varattno 5
:varatype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname o_orderdate
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:varatype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname o_orderpriority
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:varatype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname o_clerk
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
:resno 8
:resname o_shippriority
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:varatype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
:resno 9
:resname o_comment
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 9
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier false
:relid 16839
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
}

```

```

:rtkind 9
:groupexprs (
  {VAR
    :varno 1
    :varattno 6
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 6
    :location 47
  }
)
:lateral false
:inFromCl false
:securityQuals <>
}
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :reliid 10839
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 12 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
)
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
  )
}
)
:quals
{BOOLEXP
  :boolop and
  :args (
    {OPEXP
      :opno 1098
      :opfuncid 1090
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
          :varno 1
          :varattno 5
          :vartype 1082
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 5
          :location 125
        }
        {CONST
          :consttype 1082
          :consttypmod -1
          :constcollid 0
          :constlen 4
          :constbyval true
          :constisnull false
          :location 145
          :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
        }
      )
    }
    :location 137
  )
  {OPEXP
    :opno 2345
    :opfuncid 2338
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 5
        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 5
        :location 166
      }
      {OPEXP
        :opno 1076
        :opfuncid 2071
        :opresulttype 1114
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {CONST
            :consttype 1082
            :consttypmod -1
            :constcollid 0
            :constlen 4
            :constbyval true
            :constisnull false
            :location 185
            :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
          }
          {CONST
            :consttype 1186
            :consttypmod 196607
            :constcollid 0
            :constlen 16
            :constbyval false
            :constisnull false
            :location 209
            :constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 ]
          }
        )
      }
    }
    :location 198
  )
}
)
:location 178
}
{SUBLINK
  :subLinkType 0
  :subLinkId 0
  :testexpr <>
  :opetname <>
  :subselect
  {QUERY
    :commandType 1
    :querySource 0
    :canSetTag true
    :utilityStat <>
    :resultRelation 0
    :hasAggs false
    :hasWindowFuncs false
    :hasTargetSRFs false
    :hasSubLinks false
  }
}

```

```

:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname lineitem
        :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenu
          mber" "l_quantity" "l_extendedprice" "l_discount" "l_tax"
            "l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate
              "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_commen
                t")
          }
        :rtekind 0
        :relid 16566
        :inh true
        :relkind r
        :rellockmode 1
        :perminfoindex 1
        :tablesample <>
        :lateral false
        :inFromCl true
        :securityQuals <>
      }
    )
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 16566
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
      23)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
    )
    :quals
      {BOOLEXP
        :boolop and
        :args (
          {OPEXP
            :opno 96
            :opfuncid 65
            :opresulttype 16
            :opretset false
            :opcollid 0
            :inputcollid 0
            :args (
              {VAR
                :varno 1
                :varattno 1
                :vartype 23
                :vartypmod -1
                :varcollid 0
                :varnullingrels (b)
                :varlevelsup 0
                :varreturningtype 0
                :varnosyn 1
                :varattnosyn 1
                :location 325
              }
              {VAR
                :varno 1
                :varattno 1
                :vartype 23
                :vartypmod -1
                :varcollid 0
                :varnullingrels (b)
                :varlevelsup 1
                :varreturningtype 0
                :varnosyn 1
                :varattnosyn 1
                :location 338
              }
            )
          }
          :location 336
        }
      {OPEXP
        :opno 1095
        :opfuncid 1087
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 1
            :varattno 12
            :vartype 1082
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 12
            :location 365
          }
          {VAR
            :varno 1
            :varattno 13
            :vartype 1082
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 13
            :location 380
          }
        )
      }
      :location 378
    }
  }
  :location 361
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
      }
    }

```

```

:varnosyn 1
:varattnosyn 1
:location 263
}
:resno 1
:resname 1_orderkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 263
}
:resno 2
:resname 1_partkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 263
}
:resno 3
:resname 1_suppkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 263
}
:resno 4
:resname 1_linenummer
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 263
}
:resno 5
:resname 1_quantity
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 263
}
:resno 6
:resname 1_extendedprice
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location 263
}
:resno 7
:resname 1_discount
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8

```

```

:varstype 701
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location 263
}
:resno 8
:resname l_tax
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:varstype 25
:varstypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location 263
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:varstype 25
:varstypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location 263
}
:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:varstype 1002
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 263
}
:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:varstype 1002
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 263
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:varstype 1002
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 263
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:varstype 25
:varstypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location 263
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 14
:resjunk false

```

```

    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 15
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 15
    :location 263
    }
    :resno 15
    :resname l_shipmode
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 15
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 16
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 16
    :location 263
    }
    :resno 16
    :resname l_comment
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 16
    :resjunk false
    }
    }
    :location 227
    }
    :location 162
    }
    }
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
:resno 1
:resname o_orderpriority
:ressortgroupref 1
:resorigtbl 16839
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 0
:aggno -1
:aggtransno -1
:location 68
}
:resno 2
:resname order_count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 98
:sorttop 664
:reverse_sort false
:nulls_first false
:hashable true
}
)
:groupDistinct false

```

```

:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: ===== FINAL PLAN =====
DETAIL:   {AGG
:plan.disabled_nodes 0
:plan.startup_cost 194594.51589866364
:plan.total_cost 194617.5859964136
:plan.plan_rows 5
:plan.plan_width 17
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 0
:plan.targetlist (
  {TARGETENTRY
    :expr
    {VAR
      :varno -2
      :varattno 1
      :vartype 25
      :vartypmod -1
      :varcollid 100
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 0
      :varattnosyn 0
      :location -1
    }
    :resno 1
    :resname o_orderpriority
    :ressortgroupref 1
    :resorigtbl 16839
    :resorigcol 6
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {AGGREF
      :aggfnoid 2803
      :aggtype 20
      :aggcollid 0
      :inputcollid 0
      :aggtranstype 20
      :aggargtypes <>
      :aggdirectargs <>
      :args (
        {TARGETENTRY
          :expr
          {VAR
            :varno -2
            :varattno 2
            :vartype 20
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 0
            :varattnosyn 0
            :location -1
          }
          :resno 1
          :resname <>
          :ressortgroupref 0
          :resorigtbl 0
          :resorigcol 0
          :resjunk false
        }
      )
      :aggorder <>
      :aggdistinct <>
      :aggfilter <>
      :aggstar true
      :aggvariadic false
      :aggkind n
      :aggpresorted false
      :agglevelsup 0
      :aggsplit 9
      :aggno 0
      :aggtransno 0
      :location 68
    }
    :resno 2
    :resname order_count
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:plan.qual <>
:plan.lefttree
  {GATHERMERGE
    :plan.disabled_nodes 0
    :plan.startup_cost 194594.51589866364
    :plan.total_cost 194617.47599641362
    :plan.plan_rows 12
    :plan.plan_width 17
    :plan.parallel_aware false
    :plan.parallel_safe false
    :plan.async_capable false
    :plan.plan_node_id 1
    :plan.targetlist (
      {TARGETENTRY
        :expr
        {VAR
          :varno -2
          :varattno 1
          :vartype 25
          :vartypmod -1
          :varcollid 100
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 0
          :varattnosyn 0
          :location -1
        }
        :resno 1
        :resname <>
        :ressortgroupref 1
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
      }
    )
  }
}

```

```

}
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 2
:vartype 20
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 0
:varattnosyn 0
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:plan.qual <>
:plan.lefttree
{AGG
:plan.disabled_nodes 0
:plan.startup_cost 193594.49212422612
:plan.total_cost 193616.06712422607
:plan.plan_rows 5
:plan.plan_width 17
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 2
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 0
:varattnosyn 0
:location -1
}
:resno 1
:resname <>
:ressortgroupref 1
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 20
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 6
:aggno 0
:aggtransno 0
:location 68
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
)
:plan.qual <>
:plan.lefttree
{SORT
:plan.disabled_nodes 0
:plan.startup_cost 193594.49212422612
:plan.total_cost 193601.6671242261
:plan.plan_rows 2870
:plan.plan_width 9
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 3
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 6
:location -1
}
:resno 1
:resname <>
:ressortgroupref 1
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
)
:plan.qual <>
:plan.lefttree
{HASHJOIN
:join.plan.disabled_nodes 0
:join.plan.startup_cost 154455.46666666667
:join.plan.total_cost 193429.6560416667
:join.plan.plan_rows 2870
:join.plan.plan_width 9
:join.plan.parallel_aware true
:join.plan.parallel_safe true
:join.plan.async_capable false
:join.plan.plan_node_id 4
:join.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100

```

```

        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 6
        :location -1
    }
}
:resno 1
:resname <>
:ressortgroupref 1
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:join.plan.qual <>
:join.plan.lefttree
{HASHJOIN
:join.plan.disabled_nodes 0
:join.plan.startup_cost 4806.4125
:join.plan.total_cost 40362.284374999996
:join.plan.plan_rows 10210
:join.plan.plan_width 13
:join.plan.parallel_aware true
:join.plan.parallel_safe true
:join.plan.async_capable false
:join.plan.plan_node_id 5
:join.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 6
:location -1
}
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location -1
}
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
)
:join.plan.qual <>
:join.plan.lefttree
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 35383
:scan.plan.plan_rows 22585
:scan.plan.plan_width 17
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 6
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location -1
}
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location -1
}
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 3
:location -1
}
}
:resno 3
:resname <>
:ressortgroupref 0

```

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:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 4
:varatype 701
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 4
:location -1
}
:resno 4
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 5
:varatype 1082
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 5
:location -1
}
:resno 5
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 6
:varatype 25
:varitypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 6
:location -1
}
:resno 6
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 7
:varatype 25
:varitypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 7
:location -1
}
:resno 7
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 8
:varatype 23
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 8
:location -1
}
:resno 8
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 9
:varatype 25
:varitypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 9
:location -1
}
:resno 9
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:scan.plan.qual (
{OPEXPR
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4

```

```

:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 5
:location -1
}
{CONST
:consttype 1082
:constypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 145
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
)
:location 137
}
{OPEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 5
:location -1
}
{CONST
:consttype 1114
:constypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]
}
)
:location 178
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0)
:scan.plan.allParam (b 0)
:scan.scanreldid 4
}
:join.plan.righttree
{HASH
:plan.disabled_nodes 0
:plan.startup_cost 4453.25
:plan.total_cost 4453.25
:plan.plan_rows 28253
:plan.plan_width 4
:plan.parallel_aware true
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 7
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
}
:resno 1
:resname <>
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{SESCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 4453.25
:scan.plan.plan_rows 28253
:scan.plan.plan_width 4
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 8
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 5
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
}
:resno 1
:resname <>
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
}
:scan.plan.qual (
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 5

```

```

        :varattno 6
        :vartype 701
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 5
        :varattnosyn 6
        :location -1
    }
    {CONST
    :consttype 701
    :consttypmod -1
    :constcollid 0
    :constlen 8
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]
    }
    }
    :location -1
}
}
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0)
:scan.plan.allParam (b 0)
:scan.scanreldid 5
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:hashkeys (
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
)
:skewTable 16561
:skewColumn 2
:skewInherit false
:rows_total 67807.2
}
:join.plan.initPlan <>
:join.plan.extParam (b 0)
:join.plan.allParam (b 0)
:join.joinType 4
:join.inner_unique false
:join.joinqual <>
:hashclauses (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno -2
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location -1
}
{VAR
:varno -1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
)
}
:location -1
}
)
:hashoperators (o 96)
:hashcollations (o 0)
:hashkeys (
{VAR
:varno -2
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location -1
}
)
}
}
:join.plan.righttree
{HASH
:plan.disabled_nodes 0
:plan.startup_cost 135974.79166666666
:plan.total_cost 135974.79166666666
:plan.plan_rows 833461
:plan.plan_width 4
:plan.parallel_aware true
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 9
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
}
:resno 1
:resname <>

```

```

:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 135974.79166666666
:scan.plan.plan_rows 833461
:scan.plan.plan_width 4
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 10
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 325
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:scan.plan.qual (
{OEXPR
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 12
:location 365
}
{VAR
:varno 3
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 13
:location 380
}
)
:location 378
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0)
:scan.plan.allParam (b 0)
:scan.scanrelid 3
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:hashkeys (
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 325
}
)
:skewTable 16561
:skewColumn 1
:skewinherit false
:rows_total 2000306.4
}
:join.plan.initPlan <>
:join.plan.extParam (b 0)
:join.plan.allParam (b 0)
:join.jointype 4
:join.inner_unique false
:join.joinqual <>
:hashclauses (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno -2
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location -1
}
{VAR
:varno -1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0

```

```

:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 325
}
}
:location -1
}
)
:hashoperators (o 96)
:hashcollations (o 0)
:hashkeys {
  {VAR
    :varno -2
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 1
    :location -1
  }
}
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:numCols 1
:sortColIdx ( 1)
:sortOperators ( 664)
:collations ( 100)
:nullsFirst ( false)
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:aggstrategy 1
:aggsplit 6
:numCols 1
:grpColIdx ( 1)
:grpOperators ( 98)
:grpCollations ( 100)
:numGroups 5
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b)
:plan.allParam (b)
:num_workers 2
:rescan_param 0
:numCols 1
:sortColIdx ( 1)
:sortOperators ( 664)
:collations ( 100)
:nullsFirst ( false)
:initParam (b)
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b)
:plan.allParam (b)
:aggstrategy 1
:aggsplit 9
:numCols 1
:grpColIdx ( 1)
:grpOperators ( 98)
:grpCollations ( 100)
:numGroups 5
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}

```

QUERY PLAN

```

Finalize GroupAggregate (cost=194594.52..194617.59 rows=5 width=17) (actual time=1453.421..1488.327 rows=5.00 loops=1)
  Output: o.o_orderpriority, count(*)
  Group Key: o.o_orderpriority
  Buffers: shared hit=15080 read=119336, temp read=20102 written=20496
  -> Gather Merge (cost=194594.52..194617.48 rows=12 width=17) (actual time=1453.092..1488.306 rows=15.00 loops=1)
    Output: o.o_orderpriority, (PARTIAL count(*))
    Workers Planned: 2
    Workers Launched: 2
    Buffers: shared hit=15080 read=119336, temp read=20102 written=20496
    -> Partial GroupAggregate (cost=193594.49..193616.07 rows=5 width=17) (actual time=1447.205..1448.191 rows=5.00 loops=3)
      Output: o.o_orderpriority, (PARTIAL count(*))
      Group Key: o.o_orderpriority
      Buffers: shared hit=15080 read=119336, temp read=20102 written=20496
      Worker 0: actual time=1444.718..1445.651 rows=5.00 loops=1
      Buffers: shared hit=4969 read=39712, temp read=6692 written=6876
      Worker 1: actual time=1445.045..1446.047 rows=5.00 loops=1
      Buffers: shared hit=5052 read=39821, temp read=6714 written=6836
      -> Sort (cost=193594.49..193601.67 rows=2870 width=9) (actual time=1446.943..1447.402 rows=7772.33 loops=3)
        Output: o.o_orderpriority
        Sort Key: o.o_orderpriority
        Sort Method: quicksort Memory: 193kB
        Buffers: shared hit=15080 read=119336, temp read=20102 written=20496
        Worker 0: actual time=1444.491..1444.919 rows=7544.00 loops=1
        Sort Method: quicksort Memory: 193kB
        Buffers: shared hit=4969 read=39712, temp read=6692 written=6876
        Worker 1: actual time=1444.767..1445.237 rows=8288.00 loops=1
        Sort Method: quicksort Memory: 385kB
        Buffers: shared hit=5052 read=39821, temp read=6714 written=6836
        -> Parallel Hash Semi Join (cost=154455.47..193429.66 rows=2870 width=9) (actual time=1255.617..1443.558 rows=7772.33 loops=3)
          Output: o.o_orderpriority
          Hash Cond: (o.o_orderkey = linetitem_l_orderkey)
          Buffers: shared hit=15004 read=119336, temp read=20102 written=20496
          Worker 0: actual time=1262.284..1441.295 rows=7544.00 loops=1
          Buffers: shared hit=4961 read=39712, temp read=6692 written=6876
          Worker 1: actual time=1237.615..1441.466 rows=8288.00 loops=1
          Buffers: shared hit=5044 read=39821, temp read=6714 written=6836
          -> Parallel Hash Semi Join (cost=4806.41..40362.28 rows=10210 width=13) (actual time=21.368..199.311 rows=8461.00 loops=3)
            Output: o.o_orderpriority, o.o_orderkey
            Hash Cond: (o.o_custkey = c.c_custkey)
            Buffers: shared hit=4612 read=25068
            Worker 0: actual time=22.146..199.443 rows=8407.00 loops=1
            Buffers: shared hit=1524 read=8292
            Worker 1: actual time=20.623..199.504 rows=8508.00 loops=1
            Buffers: shared hit=1575 read=8488
            -> Parallel Seq Scan on public.orders o (cost=0.00..35383.00 rows=22585 width=17) (actual time=2.040..167.435 rows=18783.00 loops=3)
              Output: o.o_orderkey, o.o_custkey, o.o_orderstatus, o.o_totalprice, o.o_orderdate, o.o_orderpriority, o.o_comment
              Filter: ((o.o_orderdate >= '1994-01-01'::date) AND (o.o_orderdate < '1994-04-01 00:00:00'::timestamp without time zone))
              Rows Removed by Filter: 481217
              Buffers: shared hit=940 read=25068
              Worker 0: actual time=2.085..166.784 rows=18659.00 loops=1
              Buffers: shared hit=313 read=8292
              Worker 1: actual time=1.666..167.968 rows=18868.00 loops=1
              Buffers: shared hit=333 read=8488
            -> Parallel Hash (cost=4453.25..4453.25 rows=28253 width=4) (actual time=18.833..18.833 rows=22663.00 loops=3)
              Output: c.c_custkey
              Buckets: 131072 Batches: 1 Memory Usage: 3744kB
              Buffers: shared hit=3672
              Worker 0: actual time=18.925..18.925 rows=22204.00 loops=1
              Buffers: shared hit=1211
              Worker 1: actual time=18.658..18.658 rows=23090.00 loops=1
              Buffers: shared hit=1242
            -> Parallel Seq Scan on public.customer c (cost=0.00..4453.25 rows=28253 width=4) (actual time=0.026..13.807 rows=22663.00 loops=3)
              Output: c.c_custkey

```